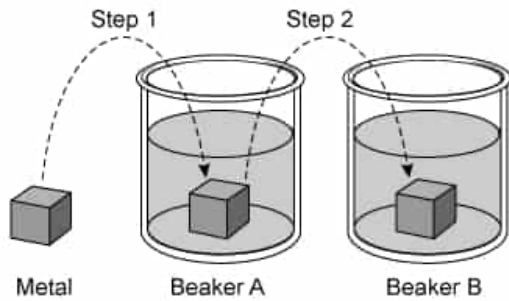




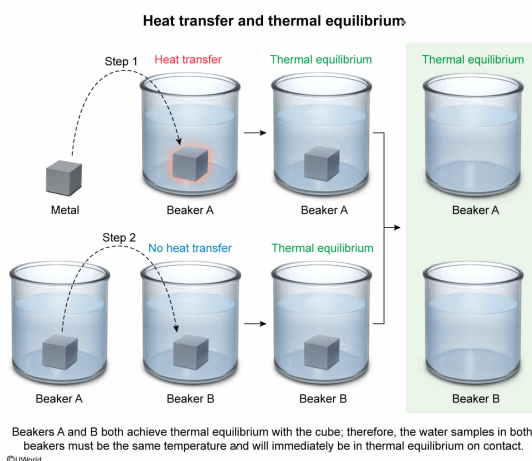
A metal cube with a temperature of 40°C is placed into 25 mL of water in Beaker A where heat is transferred to the water. After the system reaches thermal equilibrium, the metal cube is then placed into 25 mL of water in Beaker B, but no net heat transfer occurs. The cube is then removed, and the beakers are placed together so that the sides are in thermal contact. Which of the following pairs of objects are in thermal equilibrium at the end of each step of the experiment?

- I. Beaker A and the metal cube
 - II. Beaker B and the metal cube
 - III. Beaker A and Beaker B
- ☐ A. I only [7%]
☐ B. II only [4%]
☐ C. I and II only [11%]
☒ D. I, II, and III [76%]

Correct

76% answered correctly

Explanation:



At temperatures above **absolute zero**, the atoms and molecules (ie, particles) of a sample of matter are in constant motion (eg, vibration, translation, rotation) and have various amounts of **kinetic energy KE** (ie, the energy of motion). As such, a sample's **temperature is proportional to the average KE** of its particles.

When two samples are in **thermal contact**, atomic and molecular **collisions transfer energy** (ie, heat) from the sample with higher *KE* (ie, the warmer sample) to the sample with lower *KE* (ie, the cooler sample) until the particles in each sample reach **thermal equilibrium** and have the same average *KE* (ie, the same temperature). No net transfer of heat occurs between samples at thermal equilibrium.

In this question, the metal cube is initially at a higher temperature than the water in Beaker A until thermal equilibrium is established by the transfer of heat (**Number I**). When the cube is subsequently placed into the water in Beaker B, no net heat transfers which indicates that the cube and the water are already the same temperature and are immediately in thermal equilibrium on contact (**Number II**).

According to the **zeroth law of thermodynamics**, if two systems are in thermal equilibrium, and one of those systems is also in thermal equilibrium with a third system, then all three systems must be in thermal equilibrium with each other (ie, if $A \rightleftharpoons C$ and $B \rightleftharpoons C$, then $A \rightleftharpoons B$). Because the cube achieved thermal equilibrium with the water in both beakers, the water in both beakers must be the same temperature and will also be immediately in thermal equilibrium with no net heat transfer with each other on contact (**Number III**).

Educational objective:

The zeroth law of thermodynamics states that if two systems are in thermal equilibrium, and one of those systems is also in thermal equilibrium with a third system, then all three systems must be in thermal equilibrium with each other.

Time spent:0 sec

Subject:
General Chemistry

QID:403082

System:
Thermodynamics, Kinetics & Gas Laws